Cybersecurity Log Analyzer

Project Title
Cybersecurity Log Analyzer using Core Java, JDBC, and MySQL

Project Description
The Cybersecurity Log Analyzer is a menu-driven command-line application developed using Core Java, JDBC, and MySQL. It records and analyzes security-related events such as login attempts and suspicious activities, simulating a basic Security Operations Center (SOC) logging system.

Objectives
- Understand JDBC and MySQL integration
- Implement CRUD operations
- Apply layered architecture
- Simulate real-world cybersecurity logging

Technologies Used
Java – Core application logic
JDBC – Database connectivity
MySQL – Data storage
CLI – User interaction

Database Design
The project uses a MySQL database named cyberdb with a table security_logs to store log information.

Functional Modules
- Add security log
- View log by ID
- View all logs
- View failed login attempts
- Delete logs

Security Measures
- PreparedStatement to prevent SQL Injection
- Input validation in Service layer
- Proper separation of application layers

Learning Outcomes
- Practical JDBC experience
- Database design understanding
- Cybersecurity event logging concepts
- Layered architecture implementation

Future Enhancements
- User authentication system
- Password hashing
- Brute-force attack detection
- Security report generation

Conclusion
The Cybersecurity Log Analyzer successfully demonstrates how Core Java, JDBC, and MySQL can be used together to build a security-focused, real-world application following industry best practices.